

The set of natural numbers N has no identity element since 0 is excluded from the set.

5. Inverse: A set S having the identity element e with respect to a binary operation $*$ is said to have an inverse whenever, for every $x \in S$, there exists an element $y \in S$ such that: $x * y = e$.

Ex. Example: In the set of integers I with $e=0$, the inverse of an element a is $(-a)$ since $a + (-a) = 0$.

6. Distributive law: If $*$ and $\cdot$ are two binary operators on a set S , $*$ is said to be distributive over $\cdot$ whenever:

$$x * (y \cdot z) = (x * y) \cdot (x * z)$$

- An example of an algebraic structure is a field.
- A field is a set of elements, together with two binary operators, each having properties 1 to 5 and both operators combined to give property 6.
- The set of real numbers together with the binary operators $+$ and $\cdot$ form the field of real numbers.
- The operators and postulates have the following meanings:

- ① The binary operator $+$ defines addition.
- ② The additive identity is 0 .
- ③ The additive inverse defines subtraction.
- ④ The binary operator $\cdot$ defines multiplication.
- ⑤ The multiplicative identity is 1 .
- ⑥ The multiplicative inverse of $a = 1/a$ defines division, i.e. $a \cdot 1/a = 1$.
- ⑦ The only distributive law applicable is that of $\cdot$ over $+$:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c)$$

2.2 Axiomatic Definition of Boolean Algebra

In 1854 George Boole in his book "An Investigation of the Laws of Thought" introduced a systematic treatment of logic and developed for this purpose an algebraic system now called Boolean algebra.